INVESTMENT HORIZON AND INTEREST RATE RISK

Learning Outcomes

- describe the relationships among a bond's holding period return, its Macaulay duration, and the investment horizon;
- define, calculate, and interpret Macaulay duration.

We now extend the prior discussion by introducing a third investor in BRWA's 10-year, 6.2% annual eurobonds: Hightest Capital, which has an 8-year time horizon. If interest rates remain at 6.20%, the future value of reinvested coupon interest is

= –FV(0.062, 8, 6.2, 0, 0) = 61.807.

When the bond is sold, it has two years remaining to maturity. Assuming yield-to-maturity remains at 6.2% and the sale price is 100, Hightest's horizon yield (realized rate of return) is 6.20%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{61.807 + 100}{100}\right)^{\frac{1}{8}} - 1.$$

r = 0.0620 = 6.20%.

If interest rates immediately rose to 7.20% after the purchase, the future value of coupons would increase:

= –*FV*(0.072, 8, 6.2, 0, 0) = 64.071.

And the price of the bond at the time of sale would fall:

= –*PV*(0.072, 2, 6.2, 100, 0) = 98.197.

Hightest's realized eight-year horizon yield is now 6.24%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{64.071 + 98.197}{100}\right)^{\frac{1}{8}} - 1.$$

r = 0.0624 = 6.24%.

Notice that the capital loss from the bond sale of 1.803 (= 98.197 – 100.00) is roughly matched by the additional coupon reinvestment income of 2.265 from the increase in interest rates. Therefore, Hightest's total return is nearly the same as in the stable rate scenario (6.24%, compared with 6.20%).

Now assume instead that market interest rates *fall* by 100 bps, to 5.20%. The future value of coupons decreases while the price of the bond at the time of sale increases:

= –*FV*(0.052, 8, 6.2, 0, 0) = 59.630.

= –*PV*(0.052, 2, 6.2, 100, 0) = 101.854.

Hightest's realized eight-year horizon yield is now 6.17%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{59.630 + 101.854}{100}\right)^{\frac{1}{8}} - 1.$$

Rate Your Confidence
High
Medium
Low
Continue »

Category
Interest Rate Risk and Return

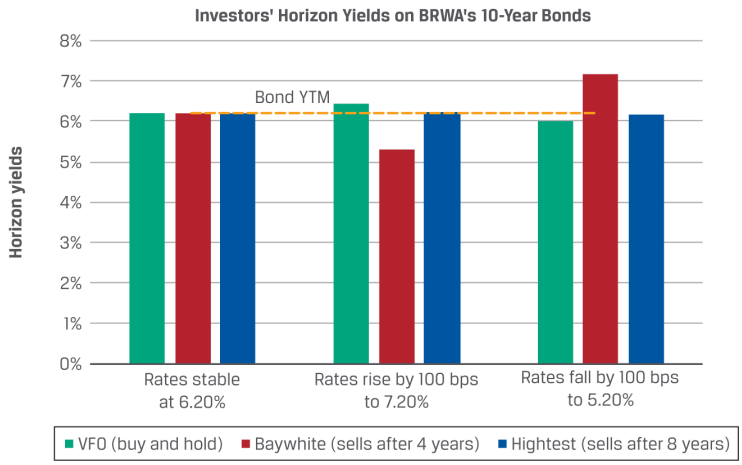
Related Questions:
Practice questions related to this topic

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

decrease in coupon reinvestment income. Again, Hightest's total return is nearly the same as in the stable rate scenario (6.17%, compared to 6.20%).

Hightest's investment results, along with those of VFO and Baywhite, are summarized in Exhibit 1. They reveal an important outcome for Hightest with the eight-year horizon: *Total returns and horizon yields are nearly the same regardless of interest rates*.

Exhibit 1



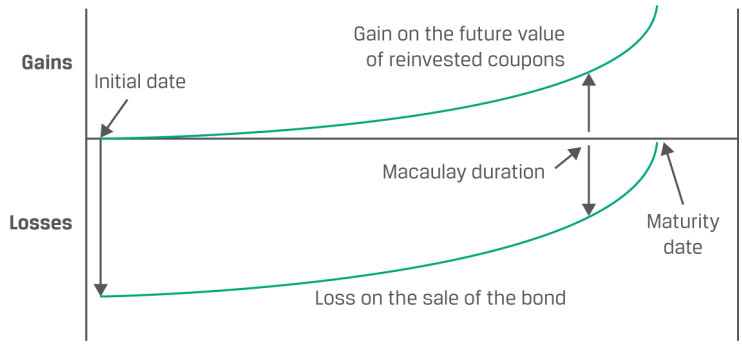
▶ Expand / collapse Extended Description

Hightest's investment horizon of eight years was chosen to illustrate an important quantity for the BRWA bond and fixed-income instruments generally: **Macaulay duration**, a holding period for a bond that balances coupon reinvestment gain (loss) and price loss (gain) for a one-time instantaneous "parallel" shift in the yield curve once the bond purchase is settled. It is named after Frederick Macaulay, the Canadian economist who introduced the concept in 1938. Given the result for Hightest, we know that the Macaulay duration of this BRWA bond is close to eight years.

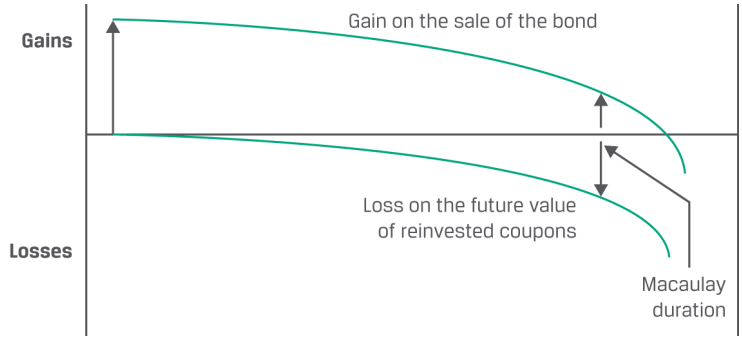
Exhibit 2 illustrates Macaulay duration for a bond initially priced at par value.

Exhibit 2:

Interest Rate Risk, Macaulay Duration, and Investment Horizon



B. Interest rates fall



As shown in Panel A, when interest rates rise, there is an immediate drop in price. Then, as time passes, the bond price is "pulled to par." The increase in the future value of reinvested coupons starts small but builds over time, and the curve indicates the added future value of reinvested coupons due to the higher interest rate. At the Macaulay duration, the two effects offset each other: the gain on reinvested coupons is equal to the loss from the increase in interest rates.

The same pattern is shown in Panel B, when interest rates fall. There is an immediate increase in price, but then the "pull-to-par" effect brings the price down as time passes. The impact from reinvesting at a lower rate starts small but becomes significant over time. Again, the Macaulay duration indicates the point when the two effects offset each other, so the gain from the decrease in interest rates equally offsets the loss from coupon reinvestment at lower rates.

The foregoing discussion allows for statements about the general relationships among interest rate risk, the Macaulay duration, and the investment horizon shown in Exhibit 3:

Exhibit 3:

Macaulay Duration, Investment Horizon, and Interest Rate Risk

	Dominant Risk	Source of Interest Rate Risk
Investment Horizon > Macaulay Duration	Reinvestment Risk	Falling Interest Rates
Investment Horizon = Macaulay Duration	Price Risk = Reinvestment Risk	—
Investment Horizon < Macaulay Duration	Price Risk	Rising Interest Rates

The Macaulay duration of the BRWA bond is 7.7429 years (the calculation will be demonstrated in the next lesson). So, Statement 1 reflects VFO's situation with its 10-year horizon; Statement 2, Hightest, the investor with the 8-year horizon; and Statement 3, Baywhite, the investor with the 4-year horizon.

Duration gap = Macaulay duration – Investment horizon. 1

VFO, with a 10-year horizon, has a negative duration gap. Therefore, the primary risk it faces is lower interest rates. Hightest, with the eight-year horizon, has a duration gap of approximately zero, so it is nearly hedged against interest rate risk. Baywhite, with the four-year horizon, has a positive duration gap and is at risk of higher rates. Importantly, as time passes, the investment horizon is reduced and the Macaulay duration of the bond also falls.

Knowledge Check

Three investors, Rook Point Investors, Fyleton Investments, and Amy Investments, purchase a 5.61%, 30-year Romanian eurobond at par value. The bond's Macaulay duration is 15.16 years. Rook Point's investment horizon is five years, while Fyleton and Amy have investment horizons of 15 and 30 years, respectively.

1. Calculate the returns for the three investors for three scenarios:

   Scenario 1: Interest rates fall by 50 bps immediately after the purchase.

   Scenario 2: Interest rates stay stable at 5.61%.

   Scenario 3: Interest rates rise by 50 bps immediately after the purchase.

   Hide Solution

   Solution:

   The total returns under the various scenarios for each investor are shown in the table below.

   The key point to note is that when an investor's investment horizon is almost equal to the bond's Macaulay duration, the returns under various interest rate scenarios remain the same. The gain (loss) from coupon reinvestment risk is offset by the loss (gain) from the price risk.

Rates Stable at 5.61%			**Rates Rise to 6.11%**			**Rates Fall to 5.11**		
Rook Point	Fyleton	Amy	Rook Point	Fyleton	Amy	Rook Point	Fyleton	A
Investment Horizon			Investment Horizon			Investment Horizo		
5	15	30	5	15	30	5	15	
MacDur > Horizon	MacDur ≅ Horizon	MacDur < Horizon	MacDur > Horizon	MacDur ≅ Horizon	MacDur < Horizon	MacDur > Horizon	MacDur ≅ Horizon	Ma Ho
Future Value of Coupons			Future Value of Coupons			Future Value of Cou		
31.379	126.765	414.223	31.694	131.678	452.199	31.067	122.063	37
Redemption Value or Sale Price			Redemption Value or Sale Price			Redemption Value or Price		
100.000	100.000	100.000	93.675	95.179	100.000	106.970	105.152	10
Horizon Yield			Horizon Yield			Horizon Yield		
5.610%	5.610%	5.610%	4.626%	5.613%	5.861%	6.659%	5.624%	5.

Question Set

Duration Gap	Risk Source
A. Negative	Rising interest rates
B. Positive	Falling interest rates

Hide Solution

Solution:

The answer to A is falling interest rates. A negative duration gap means that the bond's Macaulay duration is lower than the investor's investment horizon, which means that reinvestment risk is the dominant source of interest rate risk, from falling interest rates.

The answer to B is rising interest rates. A positive duration gap means that the bond's Macaulay duration is higher than the investor's investment horizon, which means that price risk is the dominant source of interest rate risk, from rising interest rates.

2. Complete the equation below with the following terms:

Macaulay duration

Investment horizon

Time-to-maturity

Duration gap = _____ – _____.

Hide Solution

Solution:

Duration gap = Macaulay duration – Investment horizon.

3. An investor intends to hold a bond with eight years remaining to maturity for eight years. The bond's Macaulay duration is 6.841. The investor is *primarily* exposed to which of the following interest rate risks:

A. Price risk
B. Reinvestment risk
C. Neither price nor reinvestment risk

Hide Solution

Solution:

B is correct. A buy-and-hold investor is not exposed to price risk, because she does not intend to sell the bond prior to maturity. However, her rate of return is impacted by the rate at which she can reinvest coupons.

Alternatively, the investor's investment horizon is eight years, whereas the bond's Macaulay duration is 6.841. When the investment horizon exceeds the Macaulay duration, reinvestment risk dominates.

Discuss Filter

Discussion

What do you want to discuss... (required)

SK Hello, i have attached one pratice question from this topic. Can anybody explain why SETTLEMENT date and MATURIRTY date is not considered in calculation. Thanks in advance.

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search